

Data Science Career Transition Plan

From Astrophysics Post-Doc to Munich Industry (2026–2028)

Background: PhD in Astrophysics, 10+ years post-doc experience with hydrodynamical simulations, numerical modeling, and large-scale data analysis.

Target: Data Scientist position at a Munich-based company (automotive, tech, insurance, or deep-tech startup).

Timeline: 24 months alongside current position.

Phase 1: Foundations (Months 1–6)

January 2026 – June 2026

SQL Mastery (Priority: Critical)

Your biggest gap as a physicist—industry uses relational databases, not HDF5/FITS files.

Resource	Type	Cost	Duration
Mode Analytics SQL Tutorial	Free course	€0	2 weeks
DataCamp SQL Fundamentals	Track	€25/month	4 weeks
LeetCode SQL Problems	Practice	Free	Ongoing

Milestone: Complete 50+ SQL problems, build a project querying real datasets.

Software Engineering Practices (Priority: Critical)

Move beyond Jupyter notebooks to production-quality code.

- ☐ Git workflow: branching, pull requests, code review
- ☐ Python project structure: `pyproject.toml`, virtual environments
- ☐ Testing: pytest basics, CI/CD with GitHub Actions
- ☐ Documentation: docstrings, README standards

Resource: [Real Python - Python Project Best Practices](#)

ML Refresher (Priority: Medium)

Formalize knowledge you likely already have intuitively.

- ☐ Scikit-learn official tutorials (2 weeks)
 - ☐ Hands-On ML with Scikit-Learn (Aurélien Géron) — Chapters 1-8
 - ☐ Complete 2 Kaggle “Getting Started” competitions
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Phase 2: Deep Learning & Certification (Months 7–12)

July 2026 – December 2026

Deep Learning Specialization (Priority: Critical)

Provider: DeepLearning.AI via Coursera

Cost: ~€42/month (Coursera Plus)

Duration: 3 months @ 10 hrs/week

Courses:

1. Neural Networks and Deep Learning
2. Improving Deep Neural Networks
3. Structuring ML Projects
4. Convolutional Neural Networks
5. Sequence Models

Why: Industry-standard credential, taught by Andrew Ng, highly recognized in German market.

Azure Data Scientist Associate (DP-100)

Exam Cost: ~€165

Prep Time: 6-8 weeks

Study resources:

- ☐ Microsoft Learn: DP-100 learning path (free)
- ☐ [Azure ML documentation](#)
- ☐ Practice exams on MeasureUp

Why: Azure dominates German enterprise (BMW, Siemens, Allianz all use it heavily). This is the highest-value certification for Munich market.

Phase 3: MLOps & Cloud (Months 13–18)

January 2027 – June 2027

MLOps Fundamentals

Learn to deploy and monitor models in production.

Skill	Resource	Priority
Docker	Docker for Data Science	High
MLflow	MLflow documentation	High
Kubernetes basics	Intro course only	Medium
CI/CD for ML	GitHub Actions + DVC	Medium

Cloud Platform Deep Dive

Build on Azure certification with hands-on projects:

- ☐ Deploy a model as Azure ML endpoint
- ☐ Set up automated retraining pipeline
- ☐ Implement model monitoring and drift detection

Optional: Databricks

Very popular in German enterprise—consider Databricks Lakehouse Fundamentals (free certification).

Phase 4: Job Search & Specialization (Months 19–24)

July 2027 – December 2027

Active Networking

- ☐ Attend 3+ [appliedAI Initiative](#) events (connects to BMW, Siemens, Infineon)
- ☐ Join [Data Science Munich Meetup](#)
- ☐ Attend [PyMunich](#) monthly
- ☐ Engage on LinkedIn with German market keywords

LinkedIn Optimization

Headline: Data Scientist | PhD Astrophysics | ML & Simulation Expert
Datenwissenschaftler | ML & Simulation

Summary: 10+ years analyzing terabyte-scale datasets and building numerical simulations. Transitioning from academic research to industry data science. Expert in Python, statistical modeling, and physics-informed machine learning.

Skills: Python, SQL, Machine Learning, Azure ML, PyTorch, scikit-learn, Data Visualization, Numerical Simulation

Target Companies (Munich)

Sector	Companies	Language Req.
Tech Giants	Google, Microsoft, Amazon, NVIDIA	English OK
Automotive	BMW, Audi, Continental, Infineon	Often German B2+
Deep Tech	Celonis, inovex, ClimatePartner	Usually English OK
Insurance	Allianz, Munich Re	Often German B2+
Research	Fraunhofer, Max Planck Digital	English OK

Salary Expectations (Munich)

Level	Annual Gross
Entry (career changer)	€49,000 – €57,000
Mid-level (2-3 yrs)	€57,000 – €68,000
Senior	€70,000 – €90,000+

Recruitment Agencies

- Hays Technology
 - Computer Futures
 - Michael Page
 - Robert Half
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GitHub Portfolio Strategy

Repository Structure

```
data-science-portfolio/  
  01-sql-analytics/  
    retail-sales-analysis/  
    customer-segmentation/  
  02-physics-to-industry/  
    hydrosim-predictive-maintenance/      Showcase project  
    anomaly-detection-fraud/  
    numerical-optimization-supply-chain/  
  03-deep-learning/  
    image-classification-manufacturing/  
    time-series-forecasting/  
  04-mlops/  
    azure-ml-deployment/  
    mlflow-experiment-tracking/  
  05-domain-projects/  
    automotive-sensor-analysis/  
    insurance-risk-modeling/
```

High-Impact Projects Leveraging Physics Background

Project 1: Physics-Informed Predictive Maintenance

Concept: Adapt hydrodynamic simulation methods to model industrial equipment degradation.

Tech: Python, scikit-learn, PyTorch, time-series analysis

Appeal: Automotive (BMW), manufacturing

Project 2: Cosmology Anomaly Detection → Fraud Detection

Concept: Transfer outlier detection techniques from cosmic surveys to financial transactions.

Tech: Python, imbalanced-learn, isolation forests

Appeal: Insurance (Allianz, Munich Re), fintech

Project 3: N-body Methods for Traffic Flow

Concept: Adapt particle simulation for autonomous vehicle trajectory prediction.

Tech: Python, numerical methods, visualization

Appeal: Automotive R&D

Project 4: HPC Pipeline → Cloud MLOps

Concept: Document migration from SLURM-based HPC workflows to Kubernetes/Azure ML.

Tech: Docker, Azure ML, MLflow, GitHub Actions

Appeal: Tech companies, demonstrates industry readiness

Project Checklist (Each Repo)

- ☐ Clear README with problem statement, approach, results
- ☐ Well-documented code with docstrings
- ☐ Requirements file or pyproject.toml
- ☐ Unit tests
- ☐ Visualizations of results
- ☐ Link to blog post or writeup (optional but valuable)

Certification Priority Order

Priority	Certification	When	Cost
1	Deep Learning Specialization	Month 7-9	~€125
2	Azure Data Scientist Associate (DP-100)	Month 10-12	~€165
3	Databricks Lakehouse Fundamentals	Month 14	Free
4	Optional: AWS ML Specialty	Month 16	~€300

German Language Considerations

Target Employer	Requirement
Google, NVIDIA, Celonis	English sufficient
BMW, Allianz (permanent)	German B2+ often required
Startups	Usually English OK

Recommendation: If not already proficient, invest in reaching B1-B2 level:

- ☐ Goethe-Institut course (structured)
 - ☐ iTalki tutors (flexible, conversation-focused)
 - ☐ Deutsche Welle free resources
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Monthly Checklist Template

Month: _____

Learning:

- ☐ Course/certification progress: _____
- ☐ Hours studied this month: _____

Portfolio:

- ☐ GitHub commits this month: _____
- ☐ Project milestones completed: _____

Networking:

- ☐ Meetups/events attended: _____
- ☐ LinkedIn connections made: _____
- ☐ Informational interviews: _____

Job Search (Phase 4):

- ☐ Applications submitted: _____
 - ☐ Interviews: _____
 - ☐ Feedback/learnings: _____
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Key Resources Quick Reference

Category	Resource	Link
SQL	Mode Analytics	mode.com/sql-tutorial
ML	Scikit-learn docs	scikit-learn.org
DL	DeepLearning.AI	coursera.org/deeplearning-ai
Azure	Microsoft Learn	learn.microsoft.com
Practice	Kaggle	kaggle.com
Jobs	StepStone	stepstone.de
Networking	appliedAI	appliedai.de
Meetups	Data Science Munich	meetup.com

Success Metrics

End of Year 1 (Dec 2026):

- ☐ SQL proficiency demonstrated (portfolio project)
- ☐ Deep Learning Specialization complete
- ☐ Azure DP-100 certified
- ☐ 3+ GitHub projects published
- ☐ 1 Kaggle competition completed

End of Year 2 (Dec 2027):

- ☐ 6+ polished GitHub projects
 - ☐ Active in Munich data science community
 - ☐ 50+ targeted job applications
 - ☐ Job offer secured
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Last updated: January 2026